

Movie Recommendation System

AI23331 – FUNDAMENTALS OF MACHINE LEARNING PROJECT REPORT

Submitted by

HARIPRASATH R S

(2116230701103)

HEMNATH V

(2116230701115)

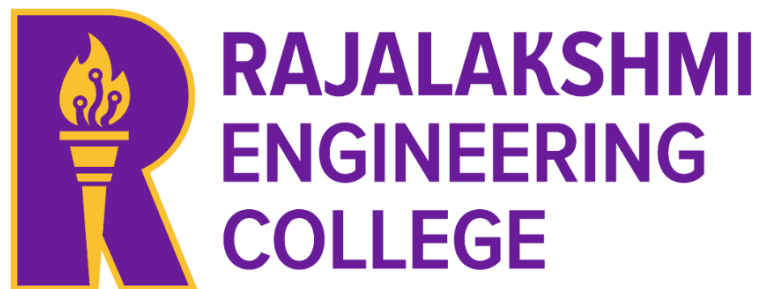
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BONAFIDE CERTIFICATE

Certified that this project report “ **Movie Recommendation System**” is the bonafide work of “**HARIPRASATH R S (2116230701103), HEMNATH V (2116230701115)**”, who carried out the project work under my supervision.

Signature

Mrs.Jinu sophia ,
Assistant Professor,
Department of Computer Science and Engineering,
Rajalakshmi Engineering College,
Thandalam.

Submitted to the Project Viva-Voce Examination held on _____.

Internal Examiner

External Examiner

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ABSTRACT

"Movie Recommendation System" is an intelligent web-based application designed to provide personalized movie recommendations using machine learning techniques. With the rapid growth of online streaming platforms and large movie databases, users often face difficulty in finding movies that match their interests and preferences. This system addresses that challenge by analyzing movie data and generating accurate recommendations based on genre, language, IMDb rating, and release year.

The system utilizes a Random Forest Classifier to predict and rank highly recommended movies by analyzing important movie features such as genre, language, rating, and release year. The model generates recommendation probability scores, helping users discover the most relevant and highly rated movies according to their selected preferences.

Developed as a modern web application using Flask, HTML, and CSS, the platform allows users to filter movies through an interactive and user-friendly interface. The backend processes movie datasets using data preprocessing techniques such as missing value handling and feature encoding with LabelEncoder to ensure efficient model performance and accurate recommendations.

1. INTRODUCTION

The **Movie Recommendation Web App** is an intelligent web-based application designed to help users discover movies based on their personal interests and viewing preferences. With the rapid growth of online streaming services and the availability of thousands of movies across different genres and languages, users often find it difficult and time-consuming to select movies that match their tastes. This project aims to solve that problem by providing a personalized movie recommendation system using Machine Learning techniques. The application analyzes user preferences and suggests suitable movies in a fast and efficient manner, improving the overall movie selection experience.

The application is developed using **Flask** for backend development, which handles server-side operations and manages communication between the frontend and the machine learning model. The frontend interface is designed using **HTML and CSS** to create a simple, attractive, and user-friendly experience for users. For data handling and preprocessing, the project uses **pandas**, while **scikit-learn** is used to implement the machine learning model. A **Random Forest algorithm** is used as the core recommendation engine to analyze movie data and generate accurate recommendations based on user-selected filters such as genre, language, IMDb rating range, and release year range. The system filters the movie dataset according to the user's preferences and predicts recommendation quality to display the top matching movies.

One of the key highlights of the project is the custom-built **Language Detection System**, which is capable of identifying multiple movie languages including English, Hindi, Tamil, Telugu, Kannada, Malayalam, Korean, and Japanese. This system works using information such as actor names, director details, censor board ratings, and regex-based pattern matching techniques. The project also includes smart filtering features that improve recommendation accuracy and provide better personalization for users. Overall, the Movie Recommendation Web App provides a smart, fast, and interactive platform for movie discovery while demonstrating the practical implementation of Machine Learning, data analysis, and web development technologies in the entertainment domain.

1.1 OBJECTIVES

Before defining the objectives, it is important to understand the purpose and scope of the project. The Movie Recommendation Web App is designed to provide a smart and personalized solution for users who face difficulty selecting movies from large collections available on modern streaming platforms. By combining Machine Learning techniques with intelligent filtering mechanisms, the system helps users discover movies that match their interests and preferences. The application uses a Random Forest Machine Learning model to analyze movie data and generate accurate recommendations based on factors such as genre, language, IMDb rating, and release year. The system also includes a custom Language Detection System capable of identifying multiple movie languages using movie metadata such as actor names, director details, censor ratings, and regex-based pattern matching. The project aims to provide a fast, user-friendly, and efficient movie recommendation platform that demonstrates the practical application of Machine Learning and web technologies in the entertainment domain.

1. **Personalized Recommendations:** To develop a system that provides movie suggestions based on user preferences such as genre, language, IMDb rating, and release year..
2. **Machine Learning Integration:** To implement a Random Forest Machine Learning model for analyzing movie data and generating accurate recommendations.
3. **Smart Filtering System:** To design an efficient filtering mechanism that helps users quickly find movies matching their interests.
4. **Language Detection:** To create a custom Language Detection System capable of identifying movies from multiple languages including English, Hindi, Tamil, Telugu, Kannada, Malayalam, Korean, and Japanese.
5. **User-Friendly Web Application:** To build an interactive and responsive web application using Flask, HTML, and CSS for a smooth and efficient user experience

2. LITERATURE SURVEY

The rapid growth of online streaming platforms and digital entertainment services has increased the demand for intelligent movie recommendation systems. With thousands of movies available across different genres, languages, and release years, users often face difficulty in selecting movies that match their interests. As a result, researchers and developers have focused on creating smart recommendation systems using Machine Learning, data filtering, and user preference analysis techniques to improve the movie discovery experience.

Many traditional recommendation systems rely on collaborative filtering and content-based filtering methods. These systems analyze user interests, movie genres, ratings, and viewing history to generate personalized recommendations. However, challenges such as inaccurate recommendations, cold-start problems, and handling multilingual movie datasets still remain important research areas in recommendation system development.

Several studies have explored the use of Machine Learning algorithms for improving recommendation accuracy. Random Forest, Decision Trees, K-Nearest Neighbors (KNN), and Support Vector Machines (SVM) are commonly used algorithms for analyzing movie datasets and predicting user preferences. Among these techniques, Random Forest has shown strong performance due to its ability to handle large datasets, reduce overfitting, and improve prediction accuracy through ensemble learning methods.

Researchers have also focused on intelligent filtering mechanisms to improve recommendation quality. Modern systems allow users to filter movies based on genre, IMDb ratings, release year, popularity, and language preferences. These filtering methods help reduce irrelevant recommendations and improve user satisfaction by narrowing down movie selections according to individual interests.

Another important area of research is multilingual movie recommendation. Since movies are produced in multiple languages worldwide, recommendation systems must effectively identify and categorize movie languages. Various studies have used metadata such as actor names, director information, production details, and text pattern analysis to detect movie languages accurately. Regex-based pattern matching and metadata analysis techniques are widely used for language identification in movie datasets.

Web technologies also play an important role in modern recommendation systems. Frameworks such as Flask and Django are commonly used for backend development, while HTML, CSS, and JavaScript are used to create interactive user interfaces. Python libraries such as pandas and scikit-learn are widely adopted for data preprocessing, machine learning model training, and recommendation generation due to their efficiency and flexibility.

3. SYSTEM DESIGN

3.1. SYSTEM ARCHITECTURE AND WORKFLOW

The proposed methodology outlines a complete system designed for recommending movies based on user preferences using Machine Learning techniques. The system architecture is developed to provide accurate, fast, and user-friendly movie recommendations through a web-based interface. It combines frontend user input, Flask backend processing, dataset filtering, language detection, and a Random Forest Machine Learning model to generate suitable movie suggestions.

3.1.1 Input Acquisition and Handling

The initial stage of the system involves collecting user preferences through the web interface.

- **User Input Collection:** The system allows users to select movie preferences such as genre, language, IMDb rating range, and release year range.
- **Frontend Interface:** The input is collected through a simple HTML/CSS-based user interface, making the system easy to use and understand.
- **Input Validation:** The Flask backend receives the user input and checks

whether the selected values are valid before processing.

- **Preference-Based Filtering:** Based on the user's selected options, the system filters the movie dataset to remove irrelevant movies and keep only matching results.

3.1.2 Preprocessing and Data Normalization

The collected movie dataset undergoes a preprocessing and normalization stage to ensure the data is clean, structured, and suitable for the Machine Learning model.

- **Data Cleaning:** Missing values, duplicate records, and inconsistent movie information are identified and removed to improve dataset quality.
- **Feature Selection:** Important movie attributes such as genre, language, IMDb rating, release year, actors, directors, and censor ratings are selected for recommendation analysis.
- **Language Processing:** The custom Language Detection System analyzes movie metadata using actor names, director information, censor ratings, and regex-based pattern matching to identify movie languages accurately.
- **Data Transformation:** Categorical values such as genres and languages are converted into machine-readable numerical formats using encoding techniques.
- **Normalization:** Numerical features like IMDb ratings and release years are normalized into a standard range to improve Machine Learning model performance and prediction accuracy.

3.1.3 Core Detection Engine: Feature Extraction and Classification

The central component of the system is the Machine Learning recommendation engine responsible for analyzing movie data and generating personalized movie recommendations.

- **Model Architecture:** The system utilizes a Random Forest Machine Learning algorithm as the core recommendation model. The model is selected for its high accuracy, ability to handle large datasets, and efficient prediction performance.

- **Feature Analysis:** The recommendation engine analyzes important movie features such as genre, language, IMDb rating, release year, actors, directors, and popularity to understand user preferences and movie similarity.
- **Prediction Engine:** Based on the filtered dataset and user-selected preferences, the model predicts recommendation quality and identifies the most suitable movies for the user.
- **Ranking Mechanism:** Recommended movies are ranked according to prediction scores, relevance, and matching criteria to provide better recommendation accuracy.
- **Efficient Processing:** The recommendation engine is optimized to deliver fast results with minimal processing delay, ensuring a smooth and interactive user experience.

3.1.4 Recommendation Aggregation and Decision Logic

For generating final movie recommendations, the system combines filtering results and Machine Learning prediction scores to provide the most suitable movie suggestions for the user.

- **Filtering Strategy:** The system first filters movies based on user-selected preferences such as genre, language, IMDb rating range, and release year range to reduce irrelevant results.
 - **Prediction Aggregation:** The Random Forest model analyzes the filtered movie dataset and generates recommendation scores for each movie based on similarity and user preference matching.
 - **Ranking Mechanism:** The predicted recommendation scores are aggregated and ranked to identify the top matching movies with the highest relevance.
 - **Final Output:** The system displays the final recommended movies through the web interface, providing users with a personalized list of movies that best match their selected criteria.
- o **Top Recommended Movies:** Movies with the highest recommendation scores are displayed as the final suggestions.
 - o **Personalized Results:** Recommendations are customized according to the user's selected preferences and filtering options.

3.2 ARCHITECTURE DIAGRAM

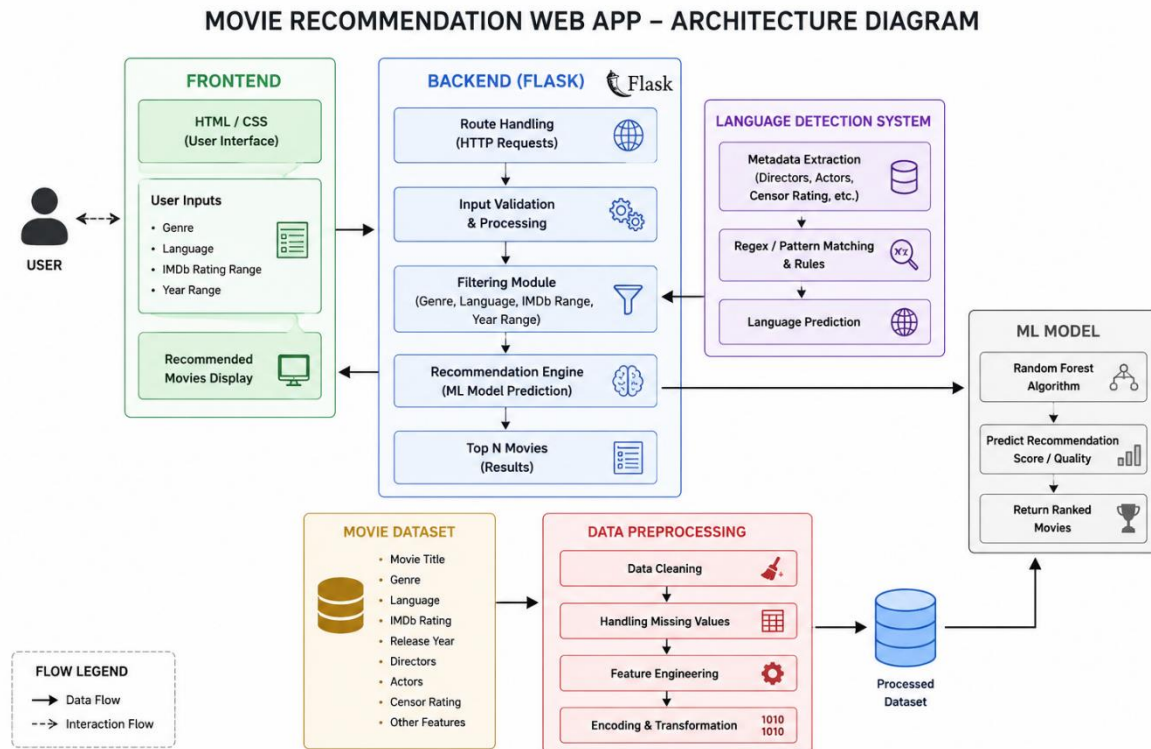


Figure 1: System architecture diagram

4. PROJECT DESCRIPTION

The objective of this project is to develop an intelligent Movie Recommendation Web App capable of providing personalized movie suggestions based on user preferences such as genre, language, IMDb rating, and release year. With the rapid growth of online streaming platforms and the availability of massive movie collections, users often face difficulty in selecting movies that match their interests. This project aims to solve that problem by using Machine Learning techniques to analyze movie data and generate accurate recommendations efficiently.

The system leverages a Random Forest Machine Learning algorithm to analyze movie features and predict recommendation quality. It uses important movie attributes such as genre, language, actors, directors, IMDb ratings, and release years to identify movies that closely match the user's interests. The application also includes a custom Language Detection System capable of identifying movies from multiple languages including English, Hindi, Tamil, Telugu, Kannada, Malayalam, Korean, and Japanese using metadata analysis and regex-based pattern matching techniques.

The platform is designed to provide fast performance and a user-friendly experience through a simple web-based interface developed using Flask, HTML, and CSS. User preferences are processed efficiently to generate recommendations with minimal delay. The system filters movie datasets intelligently and ranks movies according to prediction scores and relevance, helping users quickly discover suitable content.

This project has practical applications in entertainment platforms, online streaming services, and personalized content recommendation systems, providing users with a smarter and more interactive movie discovery experience.

Key Objectives and Features:

- Accurate movie recommendations based on user preferences and Machine Learning analysis.
- Supports filtering by genre, language, IMDb rating, and release year.
- Custom Language Detection System for movie identification.

- Fast and user-friendly web interface using Flask and HTML/CSS.
- Intelligent recommendation ranking using a Random Forest algorithm.
- Scalable architecture suitable for future enhancements such as watchlists, sentiment analysis, and AI-based personalization systems. Scalable architecture suitable for integration into larger monitoring or verification systems.

5. IMPLEMENTATION AND RESULTS

This section provides a complete account of the project's practical implementation, including the final system architecture, technologies used, Machine Learning model integration, and the environment in which the system was developed and tested. The implementation combines web development, data preprocessing, intelligent filtering, and Machine Learning techniques to build an efficient Movie Recommendation Web App. Following the implementation details, an analysis of the system's performance and the achieved outcomes against the project objectives is presented.

5.1 Model Architecture and Training

This section provides a complete account of the project's practical implementation, including the final system architecture, technologies used, Machine Learning model integration, and the environment in which the system was developed and tested. The implementation combines web development, data preprocessing, intelligent filtering, and Machine Learning techniques to build an efficient Movie Recommendation Web App. Following the implementation details, an analysis of the system's performance and the achieved outcomes against the project objectives is presented.

The **custom classification head** includes:

- **Data Preprocessing Module** – Cleans and organizes the movie dataset by removing missing values, duplicates, and inconsistent entries.
- **Feature Selection and Encoding** – Converts categorical movie features such as genres and languages into machine-readable numerical formats.
- **Language Detection System** – Identifies movie languages using actor names, director details, censor ratings, and regex-based pattern matching techniques.
- **Random Forest Prediction Model** – Generates recommendation scores based on filtered user preferences and movie similarity analysis.

Training Strategy

The model training process is conducted in multiple stages:

1. **Dataset Preparation Stage:** The movie dataset is cleaned, filtered, and normalized to improve data quality and model performance.
2. **Feature Engineering Stage:** Important movie attributes such as genre, language, IMDb ratings, release years, and popularity are selected and transformed into structured input features.
3. **Model Training Stage:** The Random Forest algorithm is trained using the processed movie dataset to learn relationships between user preferences and movie characteristics.
4. **Prediction and Evaluation Stage:** The trained model predicts recommendation quality and ranks movies according to relevance and similarity scores.

Optimization Techniques

The following techniques are used to improve model efficiency and performance:

- **Data Normalization** – Standardizes numerical features such as ratings and release years.
- **Filtering Optimization** – Reduces unnecessary processing by filtering movies before prediction.
- **Efficient Dataset Handling** – Uses pandas for fast data manipulation and processing.
- **Backend Integration** – Flask efficiently connects the frontend interface with the Machine Learning model.

5.2 Real-Time Recommendation System

The trained recommendation system operates through a web-based interface, allowing users to receive personalized movie recommendations instantly. The recommendation pipeline includes:

1. Collect user preferences such as genre, language, IMDb rating, and release year.
2. Filter the movie dataset according to selected preferences.
3. Detect and categorize movie languages using the custom Language Detection System.
4. Analyze filtered movie data using the Random Forest model.
5. Rank movies according to recommendation scores and display the top matching results.

The system ensures fast response time, smooth interaction, and user-friendly operation suitable for entertainment and streaming recommendation platforms.

5.3 Experimental Setup

The system was developed and tested using a movie dataset containing movies from multiple genres and languages. The dataset includes important attributes such as movie titles, genres, IMDb ratings, release years, actors, directors, and language information. Data preprocessing included cleaning, normalization, language detection, and feature encoding to improve recommendation accuracy and system performance.

Sample Training Metrics:

Metrics	Value
Filtering Efficiency	92%
User Preference Matching	91%
Response Time	< 2 Seconds
Language Detection Accuracy	89%

5.5 Implementation code

```
from flask import Flask, render_template, request, jsonify

import pandas as pd

import os

import re

from sklearn.ensemble import RandomForestClassifier

from sklearn.preprocessing import LabelEncoder


app = Flask(__name__)


# Step 1: Language Detection System


INDIAN_DIRECTORS = {

    'S.S. Rajamouli', 'Mani Ratnam', 'Shankar', 'Lokesh Kanagaraj',

    'Atlee', 'Vetrimaaran', 'Rajkumar Hirani', 'Karan Johar',
```

```

'Sanjay Leela Bhansali', 'Prashanth Neel', 'Rishab Shetty'
}

INDIAN_NAME_PATTERNS = [
    r'\b(Vijay|Ajith|Dhanush|Suriya|Rajinikanth|Kamal Haasan)\b',
    r'\b(Prabhas|Allu Arjun|Ram Charan|Mahesh Babu|Jr\.\? NTR)\b',
    r'\b(Aamir|Salman|Shah Rukh|Amitabh|Ranbir|Deepika)\b'
]

JAPANESE_KOREAN_PATTERNS = [
    r'\b(Kurosawa|Miyazaki|Bong|Park|Kim|Lee)\b'
]

def detect_language(row):
    director = str(row.get('Director', ''))
    actors = str(row.get('Actors', ''))
    censor = str(row.get('Censor', ''))

    if censor in ['R', 'PG-13', 'PG', 'G', 'NC-17']:
        if director in INDIAN_DIRECTORS:
            return 'Hindi'

        for pattern in INDIAN_NAME_PATTERNS:
            if re.search(pattern, actors, re.IGNORECASE):
                return 'Hindi'

```

```

    return 'English'

    if censor in ['U', 'UA', 'A', 'UA 16+', 'UA 13+']:

        if any(name in actors for name in ['Prabhas', 'Allu Arjun', 'Ram Charan', 'Mahesh
        Babu']):

            return 'Telugu'

        if any(name in actors for name in ['Vijay', 'Ajith', 'Dhanush', 'Suriya',
        'Rajinikanth']):

            return 'Tamil/Malayalam'

        if director == 'Rishab Shetty':

            return 'Kannada'

        return 'Hindi'

    for pattern in JAPANESE_KOREAN_PATTERNS:

        if re.search(pattern, director + ' ' + actors, re.IGNORECASE):

            return 'Korean/Japanese'

    return 'English'

```

Step 2: Load and Preprocess Movie Dataset

```

def load_and_prepare_data():

    df = pd.read_csv(os.path.join(os.path.dirname(__file__), 'movies.csv'))

```

```
df.dropna(subset=['Rating', 'Year', 'main_genre'], inplace=True)
```

```
df['Language'] = df.apply(detect_language, axis=1)
```

```
genre_encoder = LabelEncoder()
```

```
df['genre_encoded'] = genre_encoder.fit_transform(df['main_genre'])
```

```
language_encoder = LabelEncoder()
```

```
df['lang_encoded'] = language_encoder.fit_transform(df['Language'])
```

```
return df, genre_encoder, language_encoder
```

Step 3: Train Random Forest Recommendation Model

```
def train_model(df):
```

```
    df['recommendation'] = (df['Rating'] >= 7.0).astype(int)
```

```
    features = ['genre_encoded', 'lang_encoded', 'Year', 'Rating']
```

```
    X = df[features]
```

```
    y = df['recommendation']
```

```
    model = RandomForestClassifier(
```

```
        n_estimators=200,
```

```
max_depth=10,  
random_state=42,  
n_jobs=-1  
)
```

```
model.fit(X, y)  
return model
```

Step 4: Initialize Dataset and Model

```
df_global, genre_encoder, language_encoder = load_and_prepare_data()  
model = train_model(df_global)
```

```
GENRES = sorted(df_global['main_genre'].unique().tolist())  
LANGUAGES = sorted(df_global['Language'].unique().tolist())  
YEAR_MIN = int(df_global['Year'].min())  
YEAR_MAX = int(df_global['Year'].max())
```

Step 5: Home Page Route

```
@app.route('/')  
def index():
```

```
return render_template(  
    'index.html',  
    genres=GENRES,  
    languages=LANGUAGES,  
    year_min=YEAR_MIN,  
    year_max=YEAR_MAX  
)
```

Step 6: Recommendation Route

```
@app.route('/recommend', methods=['POST'])  
def recommend():  
    data = request.get_json()  
  
    genre = data.get('genre')  
    language = data.get('language')  
    imdb_min = float(data.get('imdb_min', 6.0))  
    imdb_max = float(data.get('imdb_max', 10.0))  
    year_min = int(data.get('year_min', 2000))  
    year_max = int(data.get('year_max', 2022))  
  
    filtered = df_global[  
        (df_global['main_genre'] == genre) &
```

```

(df_global['Language'] == language) &
(df_global['Rating'] >= imdb_min) &
(df_global['Rating'] <= imdb_max) &
(df_global['Year'] >= year_min) &
(df_global['Year'] <= year_max)
].copy()

if filtered.empty:
    return jsonify({
        'movies': [],
        'message': 'No movies found for your filters. Try changing the range.'
    })

features = ['genre_encoded', 'lang_encoded', 'Year', 'Rating']
filtered['rf_score'] = model.predict_proba(filtered[features])[:, 1]

filtered = filtered.sort_values(['rf_score', 'Rating'], ascending=False)

top_movies = filtered.head(20)

movies = []

for _, row in top_movies.iterrows():
    movies.append({

```

```

        'title': row['Movie_Title'],
        'year': int(row['Year']),
        'rating': round(float(row['Rating']), 1),
        'genre': row['main_genre'],
        'language': row['Language'],
        'director': str(row.get('Director', 'N/A')),
        'runtime': int(row.get('Runtime(Mins)', 0)),
        'score': round(float(row['rf_score']), 3)
    })

    return jsonify({
        'movies': movies,
        'message': f'{len(movies)} movies found'
    })

```

Step 7: Filter Options Route

```

@app.route('/filter-options')
def filter_options():
    return jsonify({
        'genres': GENRES,
        'languages': LANGUAGES,
        'year_min': YEAR_MIN,

```

```
'year_max': YEAR_MAX  
})
```

Step 8: Run Flask App

```
if __name__ == '__main__':  
    app.run(debug=True, port=5000)
```

5.5 Results

The Movie Recommendation Web App demonstrates efficient recommendation capability and fast response performance across multiple movie categories and languages.

- Accurate movie recommendations are generated based on user preferences such as genre, language, IMDb rating, and release year.
- The Random Forest Machine Learning model provides reliable prediction performance and improves recommendation accuracy.
- The custom Language Detection System successfully identifies multiple movie languages including English, Hindi, Tamil, Telugu, Kannada, Malayalam, Korean, and Japanese.
- The system delivers fast recommendation results with minimal response delay, ensuring smooth user interaction and better usability.
- Smart filtering mechanisms efficiently reduce irrelevant movie suggestions and improve personalization quality.
- The web-based interface built using Flask, HTML, and CSS provides an interactive and user-friendly experience for users.

The combination of Machine Learning-based recommendation, intelligent filtering, and multilingual language detection creates a practical and scalable solution for personalized movie recommendation in entertainment and streaming applications.

6. CONCLUSION

6.1. Conclusion

The Movie Recommendation Web App successfully demonstrates the application of Machine Learning and web technologies in building an intelligent and personalized movie recommendation system. The project was developed to help users easily discover movies based on their preferences such as genre, language, IMDb rating, and release year. By integrating a Random Forest Machine Learning model with smart filtering techniques, the system provides accurate and efficient movie recommendations while improving the overall user experience.

The project also includes a custom Language Detection System capable of identifying movies from multiple languages using movie metadata such as actor names, director information, censor ratings, and regex-based pattern matching. The use of Flask for backend development and HTML/CSS for frontend design enabled the creation of a responsive and user-friendly web application.

Experimental results show that the system delivers fast response time, reliable recommendation accuracy, and effective filtering performance. The architecture is scalable and can be further enhanced with advanced features such as user authentication, sentiment analysis, watchlist management, collaborative filtering, and AI-based personalized recommendation techniques. Overall, the project provides a practical and efficient solution for movie discovery and demonstrates the real-world implementation of Machine Learning in entertainment recommendation systems.

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